

Tejas Salunke

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EDUCATION

KJ Somaiya School Of Engineering

B-Tech in Computer Engineering

Vidyavihar, Maharashtra, India

Aug. 2025 – May 2029

EXPERIENCE

Computer Vision & AI Intern

Jio Platforms Limited

Ghansoli, Maharashtra, India

June 2026 – Present

- Developing and deploying advanced Computer Vision models, specializing in the integration of YOLO architectures for real-time object detection and spatial analysis.
- Architecting end-to-end AI pipelines, combining custom-trained models with existing systems to enhance accuracy and processing efficiency.
- Optimizing model inference speeds and dataset preprocessing pipelines to ensure seamless integration into production environments.

riidl Startup Growth Challenge

Competition Winner (1st Place)

Vidyavihar, Maharashtra, India

February 2026

- Awarded 1st Place for demonstrating exceptional growth strategies and executing a high-impact marketing sprint for an incubated startup.
- Implemented data-driven marketing experiments to validate product-market fit, utilizing social media outreach and funnel optimization to drive user engagement.

Somaiya Machine Learning Research Association

First Year Representative

Vidyavihar, Maharashtra, India

October 2025 – Present

- Leading student initiatives and coordinating technical workshops for a community focused on Artificial Intelligence and Machine Learning.

PROJECTS

Breast Cancer Diagnostic Pipeline | Python, PyTorch, OpenCV, NumPy, Matplotlib

June 2026 – June 2026

- Engineered a computer vision and deep learning pipeline utilizing **OpenCV** for automated tissue segmentation and artifact removal.
- Developed a binary Convolutional Neural Network (CNN) with a strict focus on clinical risk mitigation, optimizing malignant to drastically minimize dangerous false negatives.
- Implemented **Grad-CAM visual activation mapping** to explain the network's computer vision decisions, mathematically validating the morphological drivers behind detected cellular malignancy.

Genomic Sequence ML Classifier | Python, Pandas, BioPython, Scikit-Learn

June 2026 – June 2026

- Cleaned and transformed raw, unstructured FASTA genomic data into numerical matrices using k-mer extraction.
- Trained and evaluated a machine learning classifier Random Forest to identify DNA promoter regions.
- Developed an end-to-end Python pipeline, handling everything from initial data scrubbing to final predictive modeling.

Eco-Sentry: Edge AI Recycling Assistant (RAG) | LangChain, ChromaDB

March 2026 – May 2026

- Developing an Edge AI system that utilizes **Local RAG** to provide real-time, context-aware recycling instructions based on visual object detection.
- Implementing an embedded vector database using **ChromaDB** and **all-MiniLM-L6-v2** embeddings to enable high-speed local retrieval of municipal waste guidelines.
- Optimizing model performance for edge deployment by utilizing quantized **Llama 3 via Ollama**, ensuring low-latency inference without cloud dependency.

TECHNICAL SKILLS

Languages: Python, C++, Java, HTML, CSS, LaTeX

AI & Machine Learning: Deep Learning, Computer Vision, YOLO, NLP, Scikit-learn, Multi-class Classification

Generative AI & LLMs: RAG (Retrieval-Augmented Generation), LangChain, Ollama, ChromaDB

Backend & APIs: FastAPI, Django, REST APIs, API Integration

Libraries & Developer Tools: NumPy, Pandas, Matplotlib, OpenCV, Git, VS Code, PyCharm, N8N, PyQt5, Pygame